Hybrid Adaptive Transformer Differential Protection: An Intelligent DWT-LSTM Framework

1. INTRODUCTION

In the contemporary architecture of high-voltage transmission, the power transformer stands as the preeminent capital-intensive asset, the failure of which precipitates not merely localized outages but potential cascading grid instability. Within the context of a 300 MVA, 230kV/11kV transmission node, the transformer represents the cornerstone of regional energy security. The protection of this node is traditionally entrusted to the differential relay (ANSI 87T), which serves as the "Safety Backbone" by continuously supervising the current balance within the protected zone. Under healthy conditions, the 87T relay maintains the grid's equilibrium; however, as modern networks integrate increasingly complex non-linear loads and stochastic renewable transients, the reliability of this backbone is frequently tested.

The fundamental challenge in transformer protection remains the preservation of a precarious balance between security (avoidance of false trips) and dependability (guaranteed fault clearance). As grid complexity gets more intensified, classical magnitude-comparison methods tend to face diminishing returns.

1.1 Background on Transformer Differential Protection

The operational paradigm of differential protection is predicated on the "Two-Key" protocols where the primary (HV) and secondary (LV) currents are monitored at the zone boundaries. Ideally, the vector sum of these currents, while adjusted for the transformation ratio, should be zero. In the 300 MVA study case presented herein, the system utilizes a Yd1 vector group configuration, introducing a 30° phase lag on the LV side relative to the HV side.

To mitigate the resulting current mismatch, the relay should be designed to implement a mathematical compensation matrix (M_{Yd1}) to achieve phase alignment and perform "zero-sequence trapping," which will prevent maloperation during external ground faults. The compensation for the Yd1 group is defined by: $M_{Yd1} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ This matrix rotates the HV phasors by -30° and eliminates zero-sequence components. Despite such rigorous mathematical alignment, physical core phenomena introduce non-linearities that transcend classical compensation.

System Attribute	Value

Nominal Power (S_n)	300 MVA
Primary Voltage (V_{HV})	230 kV (Line-to-Line)
Secondary Voltage (V_{LV})	11 kV (Line-to-Line)
Nominal HV Secondary Current (I_{nom_hv})	37.653 A
Nominal LV Secondary Current (I_{nom_lv})	787.296 A
CT Ratio (HV & LV)	20:1
Vector Group	Yd1 (Wye-Delta 30° lag)

1.2 Magnetizing Inrush Current and CT Saturation Challenges

The "Synthetic Data Paradox" highlights the discrepancy between simulation-grade "polite" noise and the "unpredictable" physics of real-world transients. While Gaussian noise is easily filtered, the stochastic nature of inception angles and remanent flux (ϕ_0) creates severe spectral overlap between faults and benign switching events.

The Inrush Phenomenon Magnetizing inrush occurs during transformer energization, governed by the volt-second balance: $\phi(t) = \frac{1}{N} \int v(t)dt + \phi_0$. If the switching instant coincides with a critical inception angle and significant remanent flux, the core is driven beyond the 1.2 p.u. "Knee Point" into deep saturation at 1.8 p.u. flux. Consequently, the magnetizing current explodes from a negligible 0.005 p.u. to spikes reaching 5.0 p.u., a magnitude indistinguishable from internal faults to traditional relays.

CT Saturation Artifacts Current Transformers (CTs) possess their own non-linear B-H curves. During external faults with significant DC offsets, the CT core may saturate, resulting in "clipped" or "chopped" secondary waveforms. This distortion creates a "false differential" signal, acting as a distorted funhouse mirror that fools the relay into perceiving an internal fault where none exists. These phenomena necessitate a move toward waveform morphology analysis to distinguish between "rude" noise and genuine fault signatures.

1.3 Limitations of Traditional Harmonic Restraint Methods

The industry-standard "Second-Harmonic Restraint" method utilizes the R_2 energy ratio (the magnitude of the 2nd harmonic relative to the fundamental). While this can be served as a historical baseline, its efficacy is declining in modern substations. The method primarily fails due to:

1. **Weak Harmonic Signatures:** Modern transformer core materials are in most of the cases engineered for higher efficiency, which often tends to exhibit significantly lower 2nd harmonic content during inrush.
2. **CT Saturation Distortion:** Severe saturation of the CT core can suppress the spectral content of the signal, causing the harmonic signatures to vanish precisely when they are needed most.
3. **Spectral Overlap:** High-impedance internal faults or faults occurring near saturation can produce 2nd harmonic energy that causes the relay to "hang," delaying critical trip signals.

1.4 Research Problem Statement and Objectives

The central conflict in transformer protection is the "Security vs. Dependability" trade-off. The objective is to achieve 100% Precision (Security) to avoid false blackouts during inrush, while maintaining 100% Recall (Sensitivity) for internal faults, specifically those with high impedance.

Formal Research Objectives:

1. Development of a **Sliding Window Discrete Wavelet Transform (DWT)** for real-time extraction of transient spectral energy.
2. Training of an **Attention-Enhanced LSTM** network to recognize temporal signatures and "out-of-distribution" data patterns.
3. Integration of a **Hybrid "AND-Gate" Logic** combining Springer's adaptive 87T rules with AI-driven supervisory decisions to manage the mode selection priority (Inrush > CT Sat > Internal > Normal).

1.5 Proposed Hybrid Solution (DWT + LSTM + Adaptive 87T)

The proposed architecture functions as a tiered system comprising the "Nervous System," the "Brain," and the "Safety Backbone."

The Nervous System (DWT): Utilizing a "db4" (Daubechies-4) mother wavelet, a 16-sample sliding window processes current signals at 1600 Hz. Following Parseval's Theorem, the system extracts Approximation and Detail energy features ($E_{approx} = \sum(cA)^2$ and $E_{detail} = \sum(cD)^2$), totaling 6 features across all phases. This provides a refinery process that identifies "jagged" transients far more effectively than traditional frequency-ratio analysis.

The Brain (LSTM): A "Stateless ONNX" inference model processes a 32-step sliding window (20ms), providing the temporal context required for sub-cycle decision-making. The Attention mechanism ensures the model focuses on the specific millisecond of fault inception, ignoring ambient spectral leakage.

The Safety Backbone (Adaptive 87T): This layer employs Springer’s adaptive logic as a supervisory firewall. The AI does not trip independently; it permits or blocks the 87T's operation based on a confidence threshold (\ge 0.85).

Relay Mode	Identified Condition	Adaptive Settings	Conf. Count (K)
Normal	Balanced Load	Standard Slopes (S1: 0.25, S2: 0.60)	2
Inrush_Block	Magnetizing Inrush	Block Trip Output; Maintain Hold Timer	N/A
CTSat_Secure	External Fault + Sat	Increase Slope 2; Increase K	4
Internal_Trip	Internal Fault	Standard Slopes; High Confidence Permit	1-2

An **Instantaneous High-Set** override (set at 10.0 p.u.) remains active as the ultimate fail-safe, bypassing all AI logic during catastrophic, high-magnitude faults.

1.6 Thesis Structure

This thesis provides a rigorous examination of the proposed hybrid architecture across the following chapters:

- **Chapter 2: Mathematical Modeling and Physics:** A deep dive into 300 MVA transformer physics, B-H curve non-linearity, and the mathematical modeling of CT saturation.
- **Chapter 3: The DWT-LSTM Pipeline:** Detailing the "Refinery" process, moving from MATLAB-based signal processing to Python training via the ONNX bridge.
- **Chapter 4: Results and "Stress Test" Analysis:** An evaluation of system performance against the **45 \Omega high-impedance limit**, the identified "breaking point" where sensitivity begins to degrade.

- **Chapter 5: Real-time Deployment:** Integration of the trained model into Simulink for real-time protection verification and HIL readiness.

This framework represents a paradigm shift in substation automation, offering a protection scheme that is both mathematically rigorous and cognitively aware of the power system's complex transients.

THEORETICAL BACKGROUND: HYBRID ADAPTIVE TRANSFORMER PROTECTION

1. Introduction to Advanced Transformer Protection

In the engineering of high-capacity power infrastructure—specifically 300 MVA systems operating at 230 kV (Winding 1) and 11 kV (Winding 2) levels—the protection relay serves as the primary arbiter of grid stability. Traditional differential protection, while theoretically robust, faces significant operational challenges in distinguishing between catastrophic internal faults and benign transient phenomena. To maintain a strategic balance between **security**—ensuring the system does not trip during magnetizing inrush or external fault saturation—and **dependability**—the requirement to isolate actual internal faults—classical methods must be augmented with computational intelligence.

This document establishes the physical and mathematical foundation for a hybrid relaying system. By synthesizing first-principle physics with Artificial Intelligence (AI), we overcome the inherent nonlinearities of saturable transformer cores and current transformer (CT) hardware. This analysis details the transition from classical differential logic to an integrated architecture utilizing Discrete Wavelet Transforms (DWT) and Long Short-Term Memory (LSTM) networks.

2. Adaptive 87T Differential Protection

The 87T differential element is the "safety backbone" of transformer protection, operating on the principle of Kirchhoff's Current Law. However, fixed-setting relays are insufficient for 230kV/11kV infrastructure due to measurement errors and tap-changer variations; thus, an "adaptive" approach is required.

Vector Group Compensation and Normalization

For the Yd1 (Wye-Delta, 30-degree lag) transformer configuration specified in this study, mathematical alignment of the high-voltage (HV) and low-voltage (LV) currents is non-negotiable. The HV (Wye) currents must be shifted by -30° and zero-sequence components filtered to prevent maloperation during external ground faults. This is achieved via a compensation matrix: $\begin{bmatrix} I_{A,comp} \\ I_{B,comp} \\ I_{C,comp} \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix}$ Furthermore, currents are converted to per-unit (p.u.) values using nominal secondary currents ($I_{nom,hv} \approx 37.65$ A and $I_{nom,lv} \approx 787.30$ A for a 20:1 CT ratio) to ensure magnitude parity.

The Dual-Slope Characteristic

The adaptive relay evaluates the relationship between the Differential Current (I_{diff}) and the Restraint/Bias Current (I_{rest}):

- **Differential Current:** $I_{diff} = |I_{HV,comp} - I_{LV,comp}|$
- **Restraint Current:** $I_{rest} = \frac{|I_{HV,comp}| + |I_{LV,comp}|}{2}$

Tri-Zone Operation Logic

The operating characteristic utilizes three zones to maintain stability:

1. **Zone 1 (Minimum Pickup):** A threshold of 0.3 p.u. ignores measurement noise and steady-state imbalances.
2. **Zone 2 (Slope 1):** A 20–25% slope compensates for steady-state CT mismatch and tap-changer variations.
3. **Zone 3 (Slope 2):** During heavy through-faults, a steeper slope (60%) provides security against CT saturation, effectively desensitizing the relay as measurement distortion risk increases.

3. The Physics of CT Saturation

In protection simulation, a Current Transformer is often treated as a mathematical divider; however, in physical hardware, it functions as a nonlinear bottleneck.

Core Magnetization and Per-Unit Validity

The relationship between Flux and Current is defined by the B-H curve. A critical transition occurs at the "Knee Point." For the models analyzed, the saturation characteristic is [0, 0; 0.01, 10; 1, 10.5]. While the hardware utilizes a 20:1 current ratio, the saturation is defined in **per-unit (p.u.)**, rendering the saturation model valid across different hardware scales and ratios.

The Clipped Waveform Phenomenon

Exceeding the knee point causes the secondary current to no longer scale linearly with the primary. This results in:

- **Waveform Distortion:** The production of broadband high-frequency energy due to abrupt signal clipping.
- **Differential Error:** Saturation on one side of the 87T zone creates a "false" I_{diff}. The 87T logic interprets this measurement error as an internal fault, potentially triggering a false trip.

4. The Physics of Magnetizing Inrush Current

Magnetizing inrush is a benign nonlinearity occurring during transformer energization that masquerades as a catastrophic internal failure.

Derivation of the Volt-Second Balance

The fundamental relationship between applied voltage $v(t)$, turns N, and flux $\phi(t)$ is expressed via the integral form of Faraday's Law: $\phi(t) = \frac{1}{N} \int v(t) dt + \phi_0$ where ϕ_0 represents the residual flux. If the combination of the switching inception angle and ϕ_0 pushes the peak flux beyond the knee point (typically 1.2 p.u. in 300MVA cores), the magnetizing current skyrockets to 2–10 times the rated value.

Spectral Signature Comparison

Feature	Magnetizing Inrush	Internal Fault
Harmonic Content	High 2nd-harmonic (>15–20%)	Predominantly fundamental/odd
Waveform Symmetry	Significant half-cycle asymmetry	Generally symmetrical
Decay Characteristics	Temporal decay over cycles	Sustained until cleared

5. Discrete Wavelet Transform (DWT) for Transient Analysis

The DWT serves as the "optic nerve" of the hybrid relay, identifying transients that traditional Fourier analysis fails to capture.

Methodology and Feature Extraction

The implementation utilizes a **16-sample sliding window**. At a sampling frequency of **1600 Hz** within a **50 Hz** system, this 10ms window represents exactly one-half electrical cycle. The **Daubechies-4 (db4)** mother wavelet is selected for its jagged, asymmetrical shape, which optimizes the detection of fault inception spikes.

Energy Physics and Feature Mapping

The conversion of raw signals into energy features is an approximation of **Parseval's Theorem**, allowing the system to distinguish "fundamental strength" from "transient jaggedness."

1. **Decomposition:** Signals are separated into Approximation (cA) and Detail (cD) coefficients.
2. **Energy Calculation:** $E = \sum (coeff)^2$, representing the "Spectral Health" of the transformer state.
3. **Feature Mapping:** A 6-feature vector (E_{approx} and E_{detail} for phases A, B, and C) is generated every millisecond.

6. Long Short-Term Memory (LSTM) Neural Networks

If the DWT is the optic nerve, the LSTM is the "brain," capable of identifying the "story" of a fault over time.

Architecture and Recursive Hidden States

LSTMs employ "Weight Sharing," which allows a model trained on 1601-step sequences (representing 1.0 second of simulation at 1600 Hz) to remain effective during real-time inference on 32-step sliding windows. This is achieved through the **Recursive Hidden State**, which maintains temporal context across sequences.

Supervisory Role and Confidence Logic

The LSTM provides a supervisory veto to the 87T logic. It outputs a Confidence Score via a Sigmoid function (0.0 to 1.0). A threshold (λ) of 0.85 is enforced to prevent **stochastic misclassifications** from triggering false trips. The training data utilized a 3D Tensor structure [Samples, TimeSteps, Features] with the shape [1000, 1601, 6].

7. Synthesis: The Hybrid Decision Handshake

The final trip decision is a "handshake" between physical first principles and temporal pattern recognition.

Logic Truth Table: Final Trip Decision

Condition	87T State	LSTM Confidence	Action
Internal Fault	Operate	High (>0.85)	**TRIP**
Inrush Current	Operate (False)	Low	**BLOCK**
Ext. Fault (Sat)	Operate (False)	Low	**BLOCK**
High-Res Fault	Restrain (Miss)	High	**TRIP**

In the case of **High-Resistance Faults**, the hybrid system provides a significant **Sensitivity Gain**. Such faults often generate differential currents below the 0.3 p.u. 87T pickup threshold; however, the LSTM identifies the unique transient energy signatures, permitting a trip where traditional relays would remain restrained.

By integrating physical rigor with LSTM temporal analysis, this architecture ensures maximum **Security** by eliminating false trips due to inrush and saturation, while enhancing **Reliability** through the detection of low-energy internal faults.

Methodology: System Modeling and Data Generation

1. Strategic Overview of the Modeling Framework

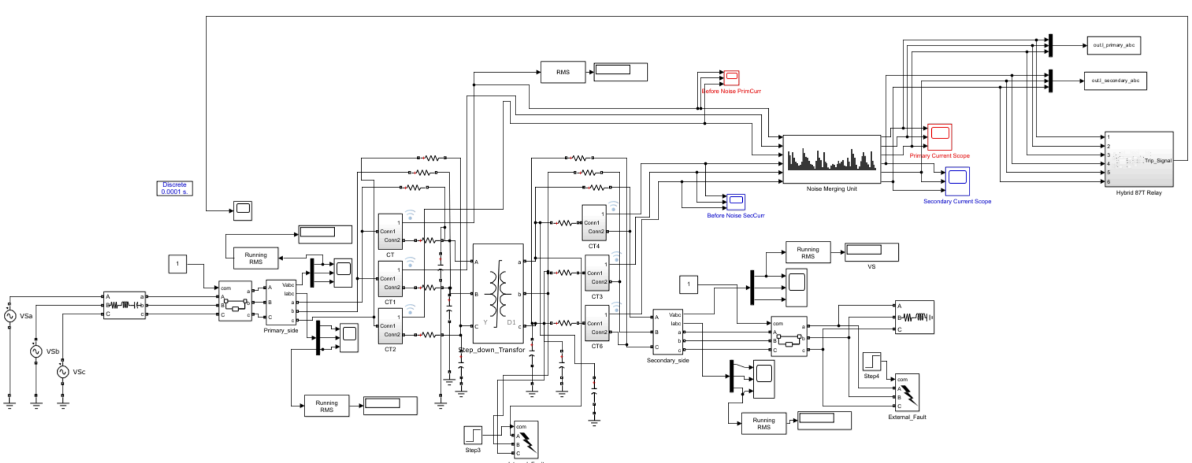
The validation of hybrid adaptive protection schemes requires a robust modeling environment capable of bridging the gap between idealized mathematical theory and the non-linear realities of power system transients. In protective relaying research, standard linear models fail to capture the nuances of magnetizing inrush and Current Transformer (CT) saturation—stochastic phenomena that frequently trigger maloperations in traditional differential relays. Consequently, a high-fidelity physical model is strategically necessary to provide a rigorous testing ground for Artificial Intelligence (AI) and Digital Signal Processing (DSP) algorithms.

The methodology follows a "Two-Step" architectural approach. First, we establish the physical power system "body" within the MATLAB/Simulink environment, incorporating transformer and CT models that exhibit realistic magnetic non-linearities. Second, we implement a stochastic data generation "engine" via MATLAB scripting. This engine orchestrates the physical model to simulate thousands of diverse operating scenarios, creating the high-dimensional data required to train the hybrid protection logic.

Crucially, the protection architecture is designed as a **Hybrid Handshake**. The LSTM does not operate as a standalone tripper; instead, it serves as a **Supervisory Veto** for the classical 87T differential logic. The relay uses a confidence threshold (\lambda = 0.85) to permit or block trips based on the AI's classification of the waveform "shape." Furthermore, an **Instantaneous Overcurrent/High-set element** (I_{high_set} = 10.0 \text{ p.u.}) is integrated as a bypass for the AI logic, ensuring that catastrophic faults are cleared with sub-cycle speed, prioritizing grid safety over classification depth.

3.1 Power System Physical Modeling (MATLAB/Simulink)

To capture high-frequency transients and non-linear magnetic behavior, we utilize a discrete-time simulation environment with a solver sample time of 1 \times 10^{-5} \text{ s} (10 \mu\text{s}). For numerical integration, we employ the **Backward Euler** solver. This choice is deliberate; while the Tustin (Trapezoidal) solver is common, it is prone to numerical oscillations or "ringing" during the abrupt transitions found in saturable models. The Backward Euler solver provides the requisite numerical stability to solve the stiff differential equations arising during non-linear magnetic excursions.



To maintain simulation stability and prevent solver conflicts—particularly "Series Inductor" errors where multiple inductive components are connected in series—we have integrated specific **Stabilization** components:

- **Three-Phase Series RLC Branches:** Small impedances (R = 0.1 \ \Omega, L = 1 \text{ mH}) are placed between the main transformer and the measurement CTs to decouple inductive components and facilitate numerical convergence.
- **Grounded Wye Load:** Because the transformer secondary is Delta-connected (floating), a grounded RLC load (10 \text{ MW}) is utilized to establish a stable 0 \text{ V} reference.
- **Resistive Snubbers:** High-resistance snubbers (R_s = 1 \times 10^6 \ \Omega) within the breaker blocks provide a leakage path during "Open Circuit" states, eliminating topological errors.

3.1.1 300 MVA, 230kV/11kV Transformer Specifications and Saturation Curve

The model centers on a 300 \text{ MVA} step-down transformer. To derive the **Per-Unit (p.u.) normalization** essential for a generalizable AI model, we first establish the foundational nameplate parameters and calculate the Full Load Amperes (FLA):

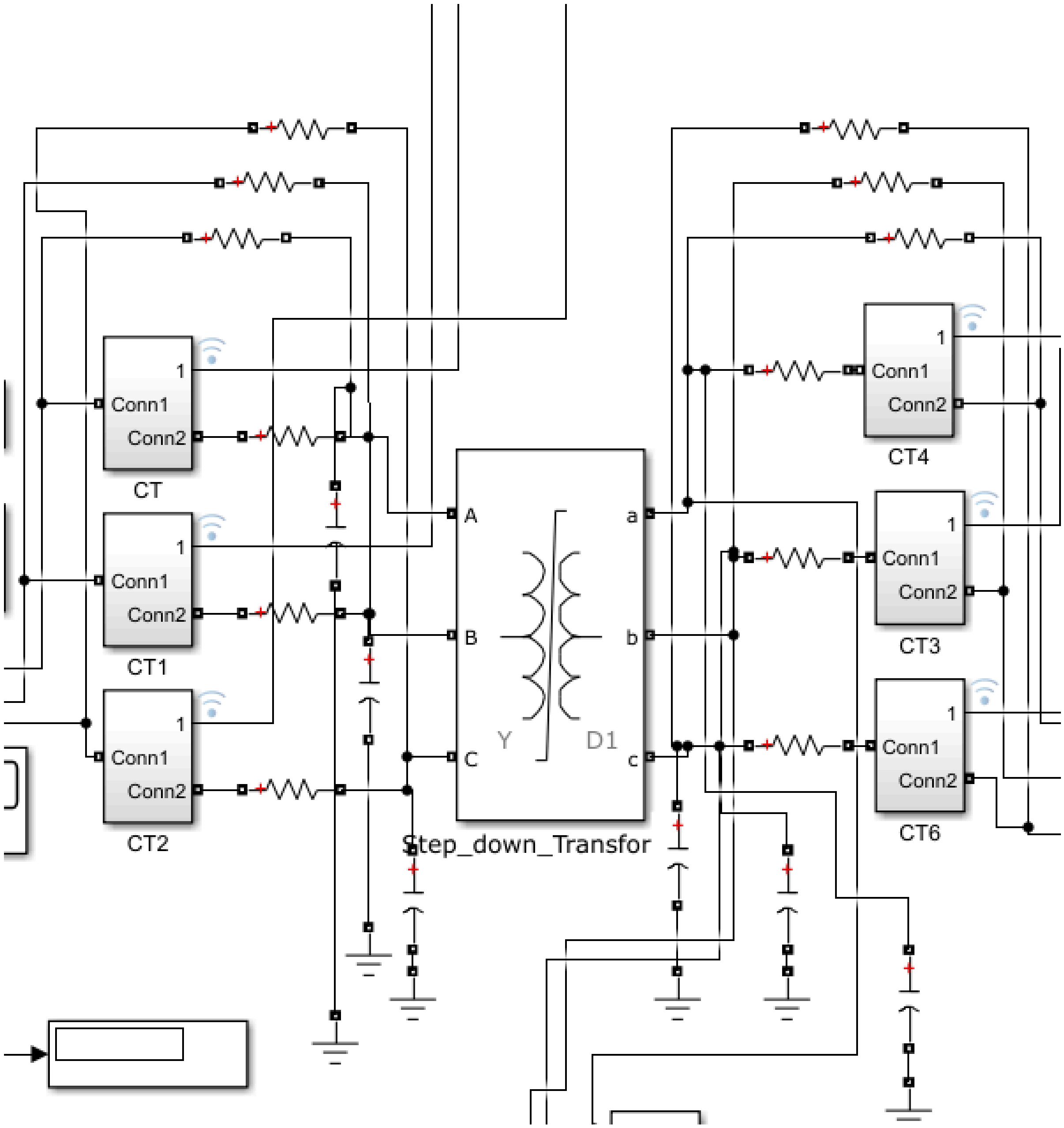
Parameter	Specification
Nominal Power (S_{nom})	300 \text{ MVA}
Frequency (f)	50 \text{ Hz}
Vector Group	\text{Yd1} (Wye/Delta, 30^\circ lag)
Winding 1 (HV)	230 \text{ kV RMS (L-L)}, R = 0.0025 \text{ p.u.}, L = 0.01 \text{ p.u.}
Winding 2 (LV)	11 \text{ kV RMS (L-L)}, R = 0.0025 \text{ p.u.}, L = 0.01 \text{ p.u.}

Magnetization Resistance (R_m)	500 p.u.
Core Type	Three single-phase units

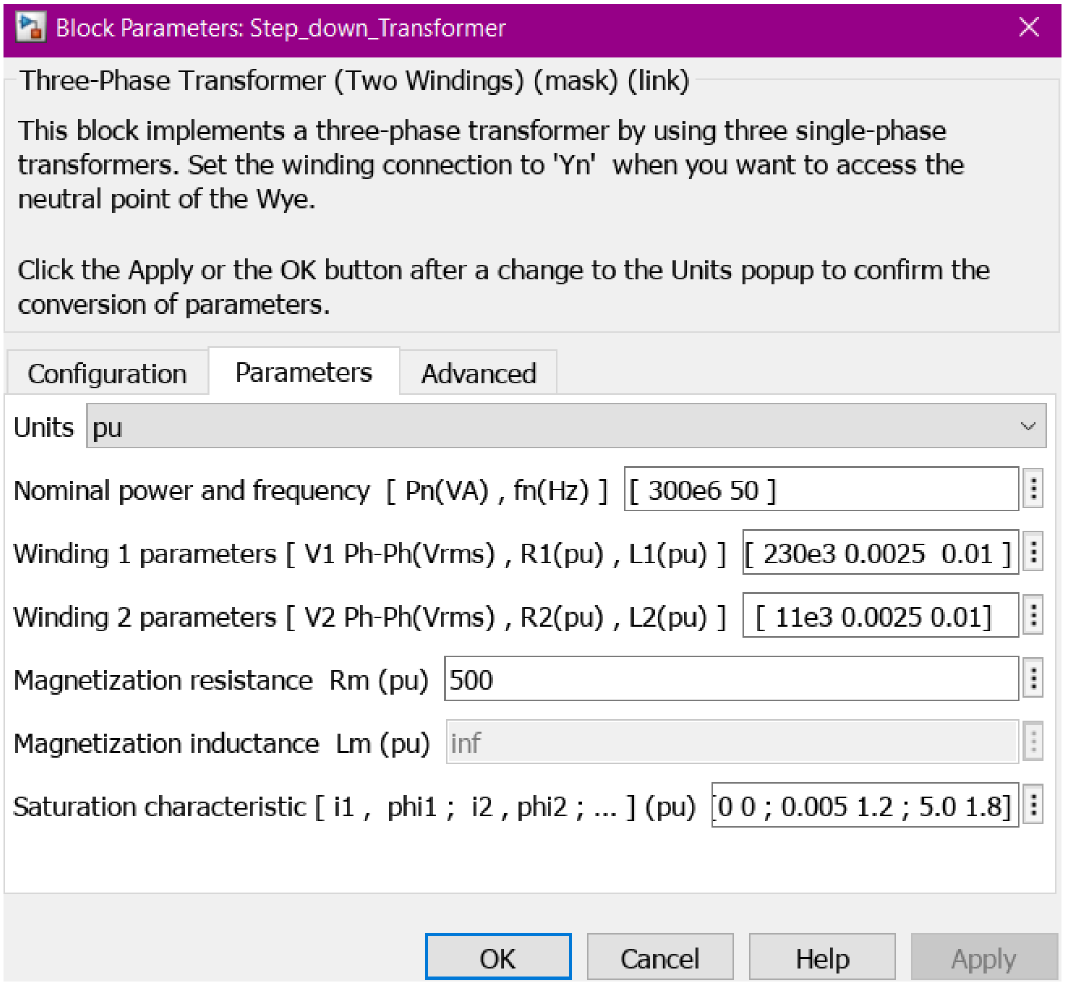
Nominal Current Calculations:

- $I_{FLA\_HV} = \frac{300 \times 10^6}{\sqrt{3} \times 230 \times 10^3} = 753.06 \text{ A}$
- $I_{FLA\_LV} = \frac{300 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = 15745.92 \text{ A}$

Because the transformer is a Yd1 vector group, the LV side lags the HV side by 30°. We implement a **Yd1 Compensation Matrix** on the HV side to align the phasors and trap zero-sequence current, a mathematical prerequisite for a secure 87\text{T} operation.



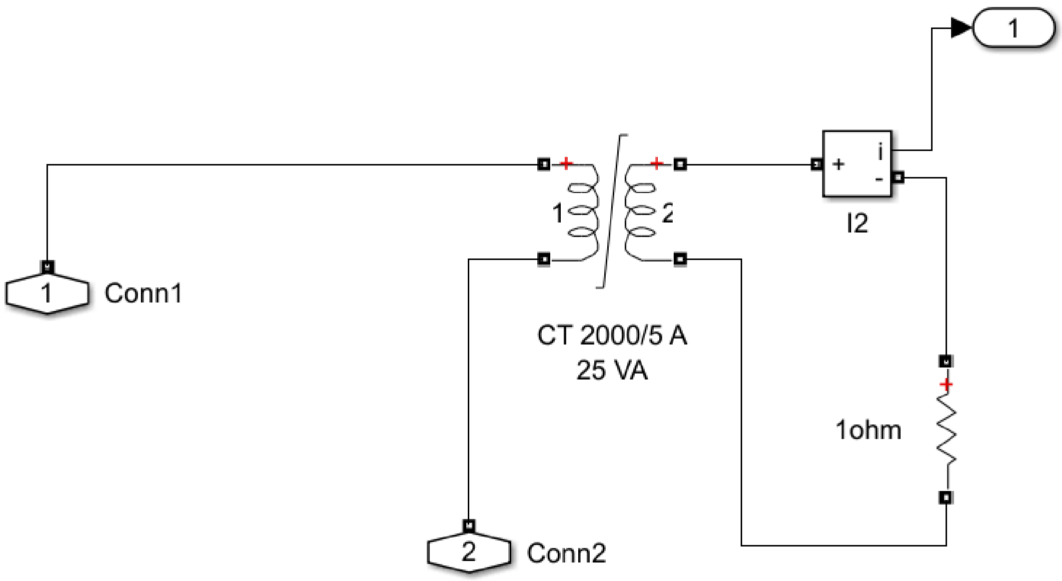
The core's transient behavior is defined by its **Saturation Characteristic** coordinates: `[0 0; 0.005 1.2; 5.0 1.8]`.



- **The Knee Point [0.005, 1.2]:** Represents the boundary of linear efficiency, requiring only 0.5\% of rated current for magnetization.
- **Deep Saturation [5.0, 1.8]:** Pushing the flux from 1.2 to 1.8 \text{ p.u.} (a 50\% increase) requires a 1000\times increase in magnetizing current (to 5.0 \text{ p.u.}). This non-linearity generates the asymmetric, harmonic-rich inrush spikes necessary for training the supervisor.

3.1.2 Current Transformer (CT) Modeling and Burden

The measurement layer utilizes CTs with a **20:1 ratio** (derived from the 0.25 \text{ V} / 5 \text{ V} winding relationship) and a 25 \text{ VA} burden capacity. Security is tested by the CT saturation curve: `[0 0; 0.01 10; 1 10.5]`. The flat characteristic after the knee point causes severe waveform clipping during heavy through-faults.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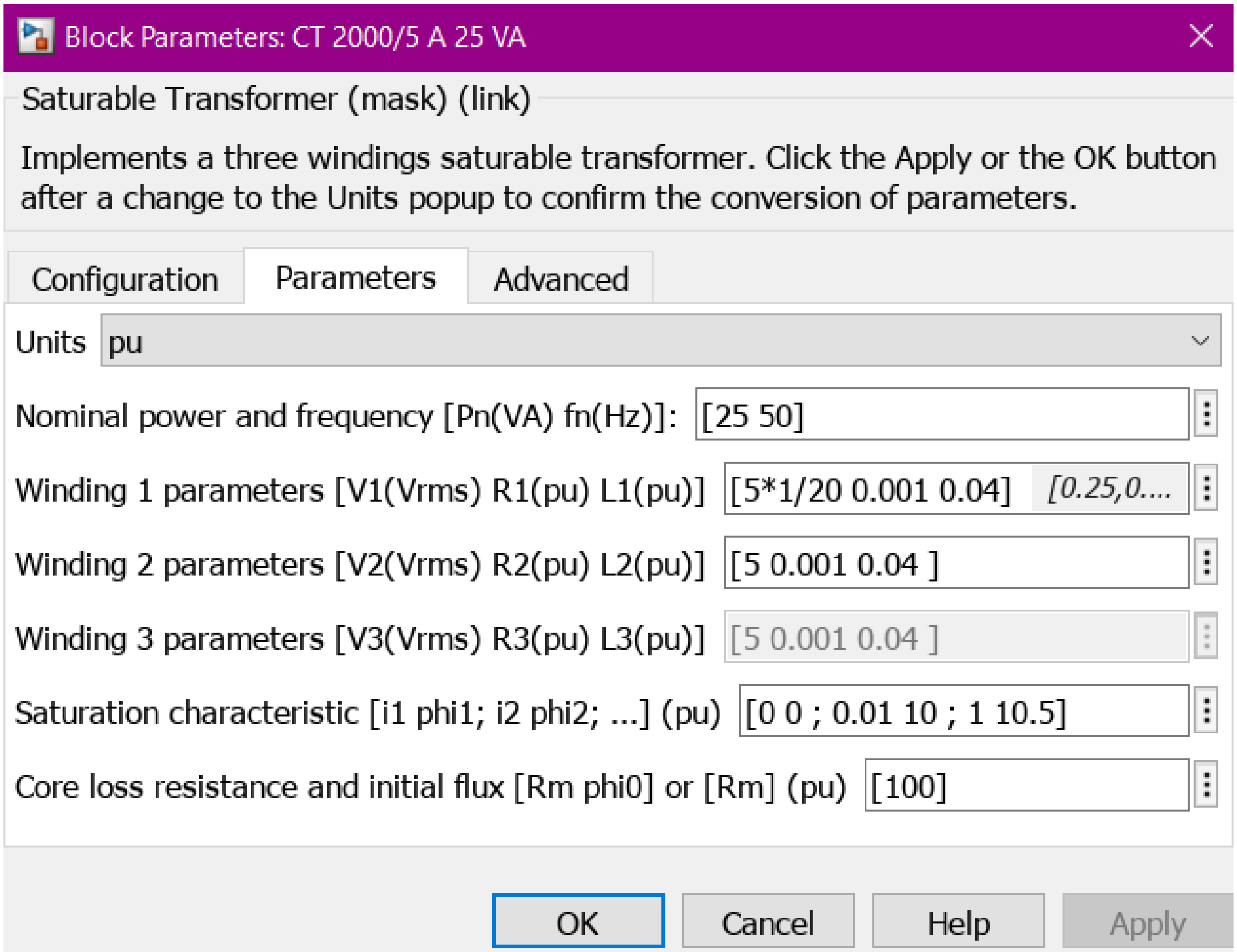
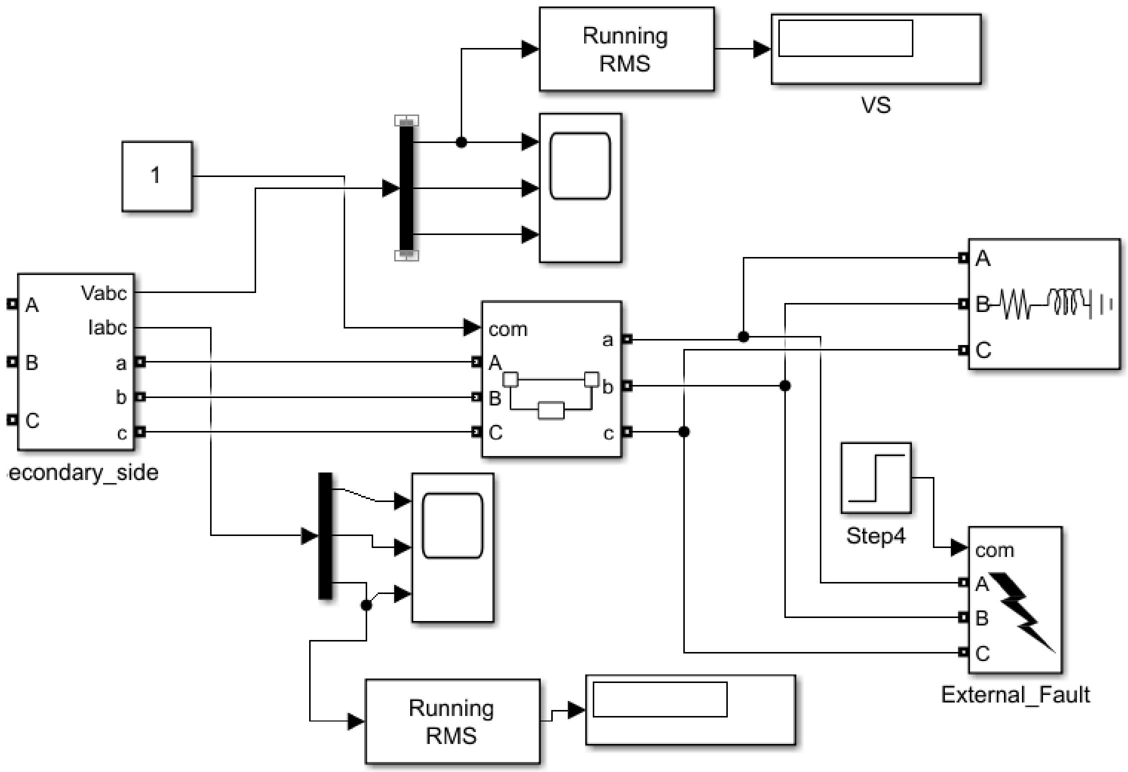


A critical stressor for the differential relay is the **Remanent Flux** (magnetic memory), which we randomize between 0.3 to 0.9 \text{ p.u.} on a per-sample basis. High remanence reduces the available flux swing before saturation, forcing the CT into the non-linear region even at lower current magnitudes. The resulting secondary imbalance creates "false" differential current, testing the relay's ability to remain secure during external events.

3.1.3 Breaker Switchgear and Snubber Resistance

Three-Phase Breaker blocks utilize "External" switching times to facilitate precise **Point-on-Wave (PoW)** energization.



By varying closure relative to the voltage phase angle, we generate worst-case inrush scenarios. The **Snubber Resistance** ($R_s = 1 \times 10^6 \ \Omega$) acts as a stability fix, providing a high-impedance path that prevents topological errors in the solver during "Open Circuit" states without introducing significant leakage current.

3.2 Signal Conditioning and Hardware Emulation (The Merging Unit)

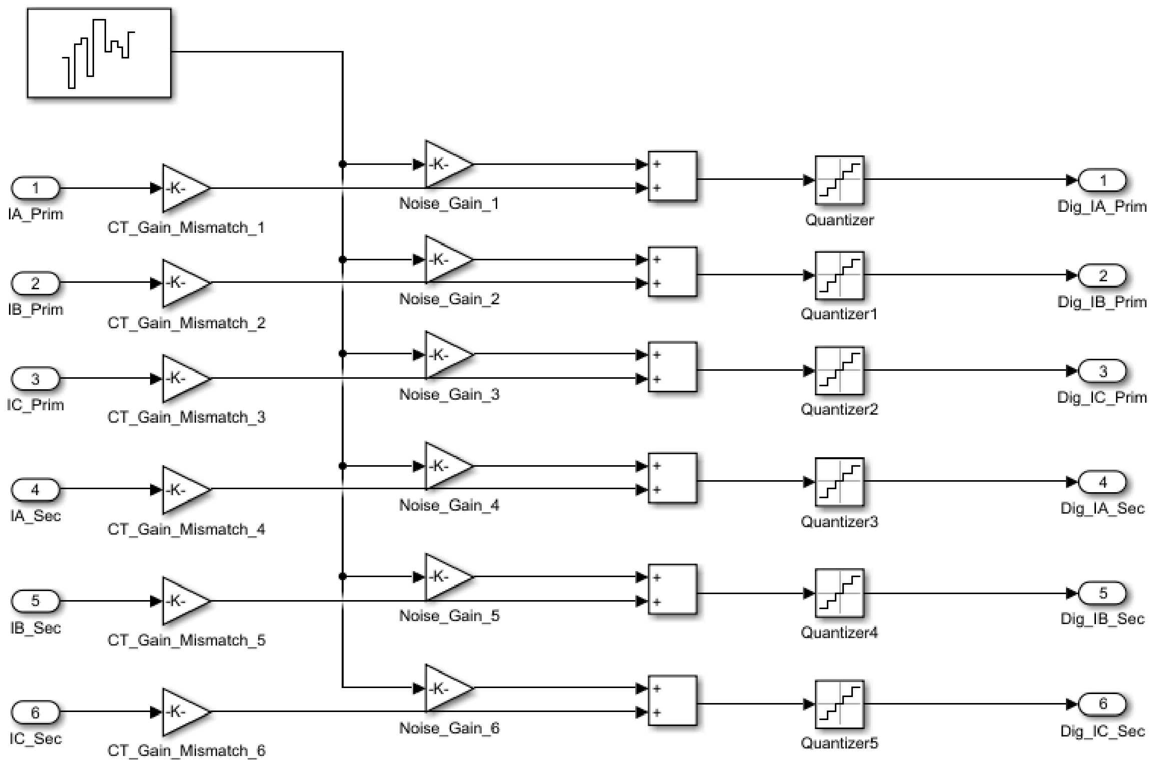
The Merging Unit acts as the "Nervous System," transforming raw electromagnetic signals into digitized packets. While the physical solver runs at $10 \ \mu\text{s}$, the relay acquisition layer samples the data at **1600 Hz** (0.625 ms interval), resulting in 1601 discrete steps for a 1.0 s simulation.

3.2.1 DWT Engine and Feature Extraction

The digital relay implements a **Discrete Wavelet Transform (DWT)** engine for multi-resolution analysis (MRA). Using a **db4 (Daubechies 4) mother wavelet**, chosen for its jagged shape that matches fault inception transients, the system processes a **16-sample (half-cycle) sliding window**.

Applying **Parseval’s Theorem**, we approximate the energy within the Approximation and Detail coefficients. This produces a 6-dimensional feature vector for every time step (t): $V_t = [E_{approx\_A}, E_{detail\_A}, E_{approx\_B}, E_{detail\_B}, E_{approx\_C}, E_{detail\_C}]$ This spectral summary tells the AI whether the signal is a smooth fundamental sine wave (high E_{approx}) or a jagged transient/noise (high E_{detail}).

3.2.2 Impairment Injection: Noise and Quantization



To ensure robustness, the Merging Unit introduces real-world imperfections:

- **CT Mismatch:** Gain error of up to 2% is injected, necessitating the "Slope 1" bias in the 87T logic.
- **AWGN:** Band-Limited White Gaussian Noise is applied with SNRs ranging from **30dB** (standard) to **15dB** (extreme). High-frequency noise mimics the detail coefficients of a fault, testing the LSTM’s ability to differentiate stochastic noise from deterministic transients.
- **16-Bit ADC Quantization:** Continuous signals are quantized to mimic the finite precision of digital hardware like the SEL-787.

3.3 Automated Batch Data Generation (ControlPanel.m)

The `ControlPanel.m` script automates a Monte Carlo-style generation of 1,000+ operating scenarios, ensuring the AI model is generalizable and avoids the **"Synthetic Data Paradox"**—where models perform perfectly in clean simulations but fail in the noisy field.

3.3.1 Normal, Inrush, and Fault Scenarios

The script distinguishes between several classes of operation:

- **Normal Load:** Light loading ($10 \text{ MW}, \approx 3\%$).
- **Magnetizing Inrush:** Triggered by breaker closure with randomized residual flux. The script captures the decaying envelope, second-harmonic energy, and half-cycle asymmetry.
- **Internal vs. External Faults:** Distinguishes between faults within the protected zone (`shouldTrip = 1`) and through-faults with CT saturation (`shouldTrip = 0`).

3.3.2 Stochastic Parameter Randomization

To cover the fault space comprehensively, parameters are randomized:

1. **Fault Resistance (R_f):** 0.01 to $20 \ \Omega$ for standard training; 50 to $100 \ \Omega$ for high-impedance stress testing.
2. **Inception Angle:** 0° to 360° to capture faults at voltage peaks and zero-crossings.
3. **Fault Type:** All permutations (AG, BG, CG, AB, BC, CA, ABC, ABG, etc.).
4. **Remanent Flux:** 0.3 to 0.9 p.u. randomization per sample to test the limits of CT saturation security.

By utilizing DWT energy as the input, we provide the LSTM with "pre-solved" feature separation—acting as a "flare in a dark room" that allows the neural network to differentiate complex magnetic distortions from actual winding faults with high accuracy.

Chapter 4: Data Preprocessing and AI Model Development

The efficacy of an intelligent transformer protection system is fundamentally governed by the integrity of the data ingested by its deep learning architectures. In the evolution from analog electromagnetic relays to digital signal processing (DSP) environments, the strategic acquisition of high-fidelity data is the prerequisite for establishing a reliable "ground truth." This chapter details the transition from raw, continuous analog current signals to discrete digital samples, forming the basis for a hybrid model capable of distinguishing between benign magnetizing inrush and catastrophic internal faults.

4.1 Data Extraction and Digital Sampling

The initial stage of the preprocessing pipeline involves the precise conversion of raw physical signals into a discrete format. This process must preserve high-frequency transients and non-linearities, such as those caused by CT saturation and Yd1 vector group phase shifts, to ensure the secondary feature extraction stage can isolate hidden fault signatures.

4.1.1 1.6 kHz Discrete Sampling Rate (32 samples/cycle)

For a 50 Hz power system, a sampling frequency of 1.6 kHz is utilized to achieve a resolution of exactly 32 samples per cycle (1600 / 50 = 32). Within a 1.0-second simulation window, this produces a total of 1601 samples (1600 steps plus the initial t_0 condition).

The technical justification for selecting 1600 Hz over higher industrial rates involves several strategic considerations:

- **Transient Capture:** This rate is sufficient to capture the jagged discontinuities of CT saturation and high-frequency fault inception spikes.
- **DC Offset Preservation:** The resolution allows the model to accurately track the decaying DC offsets characteristic of magnetizing inrush, which are vital for differential logic stability.
- **Computational Latency:** 1.6 kHz represents an optimal trade-off between signal granularity and real-time processing constraints. Lower rates risk aliasing critical transients, while higher rates impose unnecessary computational overhead on the relay's processor, potentially delaying trip decisions beyond the critical sub-cycle window.

4.1.2 Quarter/Half-Cycle Sliding Window Implementation

To mimic the operational behavior of a real-time digital relay, a sliding window methodology is implemented. Rather than analyzing a full second of data offline, the system processes snapshots of 16 samples, representing a 10ms (half-cycle) temporal window. At every discrete time step (t), the window shifts forward by exactly one sample—a mechanism known as "The Slide."

This implementation forces the model to prioritize local transient behavior over average event signatures. While the model is trained on longer 1601-step sequences, the use of a 32-step sliding window (20ms) during deployment creates a "Time Machine" effect. This allows the protection logic to make high-confidence decisions within milliseconds of fault inception, as the LSTM tracks the evolution of the signal across sequential 10ms snapshots. This weight-sharing approach over time ensures that the model recognizes the "story" of a fault as it unfolds, rather than waiting for a full cycle to conclude.

4.2 Feature Engineering via Discrete Wavelet Transform (DWT)

Raw sinusoidal waveforms are inherently insufficient for training Long Short-Term Memory (LSTM) networks in power systems; the fundamental 50 Hz component changes too slowly, often leading to vanishing gradients or model confusion during inrush events. The Discrete Wavelet Transform (DWT) is employed as a "spectral lens" to isolate hidden fault signatures and discontinuities that are obscured in the time-domain.

4.2.1 Daubechies-4 ('db4') Wavelet Decomposition

The 'db4' mother wavelet is selected for signal decomposition due to its asymmetrical and jagged shape. This is the industry standard for matching the unique morphology of power system transients and fault inception spikes. Through a single-level decomposition, the signal is partitioned into two distinct coefficient sets:

1. **Approximation Coefficients (cA):** These capture low-frequency components, specifically the fundamental 50 Hz signal and the slowly decaying DC offset.
2. **Detail Coefficients (cD):** These isolate high-frequency "snaps," CT saturation noise, and the immediate spectral discontinuities of a fault.

4.2.2 Approximation and Detail Energy Calculation

Utilizing principles of Parseval’s Theorem, these coefficients are transformed into quantifiable energy features. This transformation summarizes "spectral health" by squaring and summing the coefficients: $E = \sum (coefficients)^2$.

Phase	Feature	Significance
Phase A	E_{approx_A}	Fundamental frequency health; detects loss of phase current.
Phase A	E_{detail_A}	Transient discontinuities; primary indicator of fault inception.

Phase B	E_{approx_B}	Fundamental energy; identifies phase-specific magnitude drops.
Phase B	E_{detail_B}	High-frequency signatures; differentiates inrush from faults.
Phase C	E_{approx_C}	Low-frequency baseline for Phase C.
Phase C	E_{detail_C}	Captures asymmetrical fault spikes and saturation noise.

Converting 16 raw data points into two energy features per phase significantly reduces dimensionality while increasing information density. High E_{detail} energy serves as the primary indicator of fault inception or severe CT saturation. By training the AI on these energy sequences, it learns that smooth sine waves yield low E_{detail}, while jagged fault transients yield distinct energy bursts.

4.2.3 3D Tensor Formatting and Dimension Permutation

For ingestion into the PyTorch training pipeline, data is structured as a 3D Tensor with dimensions [1000, 1601, 6]. These indices represent 1000 scenarios, 1601 time steps, and 6 energy features (2 per phase).

A critical technical requirement is addressing the "Column-Major" (MATLAB) vs. "Row-Major" (Python) storage mismatch. When loading the processed files (e.g., `Processed_LSTM_Data_20260119_120000.mat`), the data must be permuted using the `.permute(2, 1, 0)` operation. This shifts the indices from (6, 1601, 1000)—where features are the first dimension—to the (1000, 1601, 6) "batch-first" format required by PyTorch. Failure to execute this permutation causes the model to confuse sample counts with feature counts, leading to total model divergence.

4.3 LSTM Architecture and Training Pipeline

The LSTM network serves as the supervisory intelligence of the hybrid relay. Its gated memory cells allow it to "remember" the specific sequence of energy spikes, enabling it to differentiate between the harmonic decay of a magnetizing inrush and the persistent transients of an internal fault.

4.3.1 Model Hyperparameters and Global Temporal Attention

The architecture utilizes 128 hidden units across 2 layers with a Dropout rate of 0.3 to ensure robust generalization. To solve the computational bottleneck associated with long sequences (1601 steps), a **Global Temporal Attention Mechanism** is integrated.

Because a fault inception may only last 2–5ms within a 1000ms window, the Attention Mechanism forces the network to focus exclusively on the high-energy transient milliseconds while ignoring the pre-fault steady state. This ensures that the LSTM identifies the specific "moment of truth" during the event, rather than being confused by the average energy of the entire sequence.

4.3.2 Validation Framework and K-Fold Cross-Validation

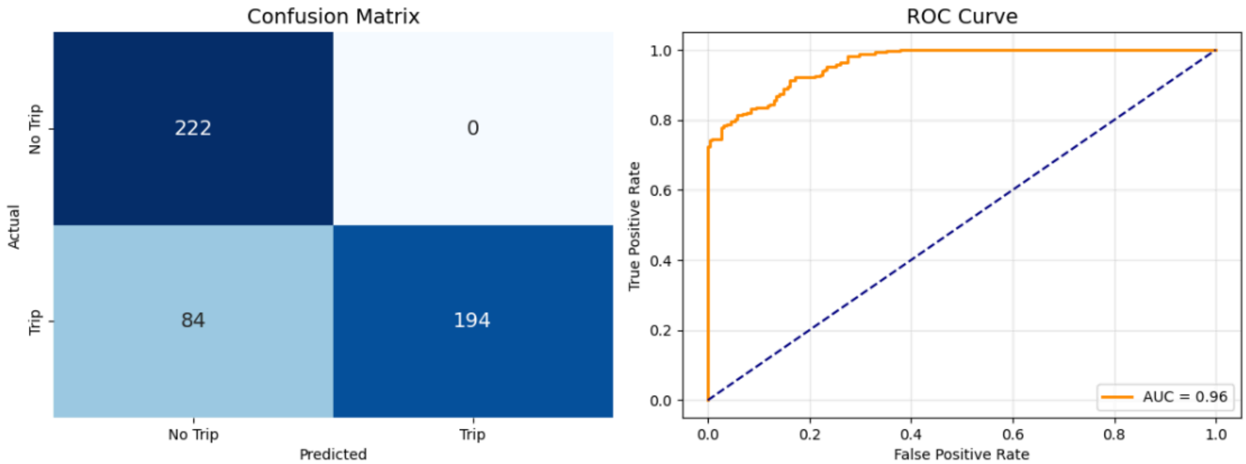
Model generalization is ensured via an 80/20 split (800 training/200 testing samples) and a rigorous **5-Fold Cross-Validation** protocol. By rotating the data through five separate folds, we prove the model's robustness against varying fault resistances (R_f), inception angles, and remanent flux levels. Data is managed using `TensorDataset` and `DataLoader` with a batch size of 32 to maintain gradient stability during GPU-accelerated training.

4.3.3 Optimization and the "Stress Test" Requirement

The training loop utilizes the **Adam Optimizer** and **Binary Cross Entropy (BCE) Loss**, supplemented by a `ReduceLROnPlateau` scheduler and `Early Stopping` to prevent overfitting. In the context of protection engineering, performance is evaluated through a safety-critical lens:

- **Recall (Safety):** The ability to never miss a real fault. A false negative could result in transformer destruction.
- **Precision (Stability):** The ability to avoid false trips during CT saturation or inrush. A false positive can lead to cascading grid blackouts.

While the model may achieve near 100% accuracy on standard datasets, such results are often a byproduct of "polite" Gaussian noise. To validate the relay for real-world deployment, it must undergo a **Stress Test** using "OOD" (Out-of-Distribution) data, including High Impedance Faults (up to 100 \Omega), extreme 15dB noise levels, and 80% CT remanent flux mismatches.

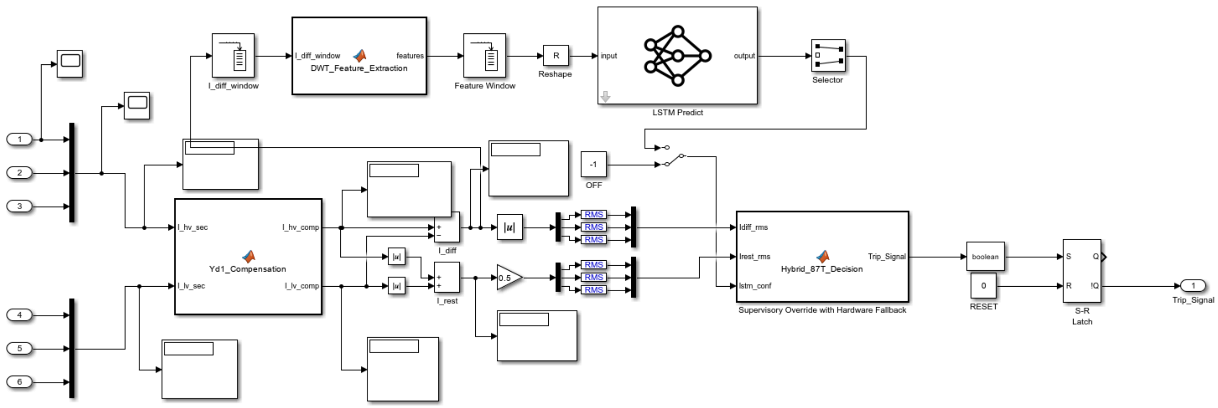


This integrated pipeline—from 1.6 kHz sampling to the high-confidence LSTM output—enables the final hybrid decision logic. The LSTM acts as a supervisory "Veto" or "AND gate" for the classical adaptive 87T logic; the relay only trips if the physical differential laws are violated and the AI confirms the internal fault signature with high confidence.

Real-Time Implementation of a Hybrid DWT-LSTM Relay for Transformer Protection

5. HYBRID RELAY REAL-TIME IMPLEMENTATION (SIMULINK)

The transition from offline validation to online real-time protection constitutes the "final frontier" in verifying the DWT-LSTM architecture's efficacy. While historical datasets confirm the model's pattern recognition capabilities, the dynamic power system environment of a 300MVA transformer introduces stochastic transients, point-on-wave switching variations, and frequency deviations that challenge even the most robust architectures. Real-time implementation in Simulink serves as the ultimate laboratory for sub-cycle clearing performance, ensuring the AI-driven logic operates within the deterministic constraints required for modern grid security.



5.1 Real-Time Signal Pre-Processing

Signal pre-processing acts as the "nervous system" of the relay, providing the necessary galvanic isolation and anti-aliasing filtering before the raw physical transients are ingested by the logic engines. For a 300MVA transformer, measurements must be meticulously conditioned to account for vector group phase shifts and hardware mismatches in the secondary burden of the 20:1 current transformers (CTs).

5.1.1 Yd1 Vector Group Phase Alignment and Zero-Sequence Trapping

A fundamental requirement for protecting a Yd1-connected transformer is the mathematical alignment of the HV Wye and LV Delta currents. The Delta side naturally lags the Wye side by 30°, which would manifest as a massive false differential current under normal load if left uncompensated. Furthermore, the Wye side is susceptible to zero-sequence currents during external ground faults—components that do not circulate in the Delta winding. To prevent maloperation, the HV secondary currents are processed through a compensation matrix to shift them by -30° and "trap" zero-sequence components.

The compensation operation is formulated as:

$$\begin{bmatrix} I_{A(comp)} \\ I_{B(comp)} \\ I_{C(comp)} \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} I_{A(hv)} \\ I_{B(hv)} \\ I_{C(hv)} \end{bmatrix}$$

5.1.2 Secondary Nominal Current (I_{nom}) Per-Unit Normalization

Due to the vast disparity between the 230kV HV and 11kV LV windings, the raw secondary currents differ significantly in magnitude despite identical 20:1 CT ratios. To enable universal evaluation across the DWT-LSTM and 87T engines, the relay operates in the Per-Unit (p.u.) domain. The normalization constants (Taps) are derived as follows:

1. **Calculate Primary Full Load Amps (FLA):** Based on the 300MVA rating, FLA_{hv} is 753.06 A and FLA_{lv} is 15745.92 A.
2. **Determine Secondary Nominal Values:** By applying the 20:1 CT ratio, we derive:
   - $I_{nom\_hv} = 37.653$ A
   - $I_{nom\_lv} = 787.296$ A
3. **Normalization:** All live secondary measurements are divided by these constants to reach a 1.0 p.u. base.

5.1.3 Differential (I_{diff}) and Restraint (I_{rest}) RMS Calculation

The hybrid architecture employs a bifurcated data pipeline. The LSTM engine ingests raw, high-resolution DWT transients to identify spectral health, while the classical 87T logic requires stable magnitudes for its dual-slope operate regions. To avoid "chatter" in the trip flags during zero-crossing events, a "Sliding Window RMS" filter is implemented for the 87T logic. This ensures that I_{diff} and I_{rest} remain steady magnitudes during sustained faults, whereas the DWT coefficients capture the jagged high-frequency discontinuities indicative of fault inception or CT saturation.

This signal conditioning ensures that both the deterministic physical math and the stochastic neural inference receive data optimized for their respective decision-making processes.

5.2 Neural Network Inference Integration

The deployment of deep learning within a time-critical protection chain is facilitated by the Open Neural Network Exchange (ONNX) format. ONNX serves as the "Universal PDF" for machine learning, enabling PyTorch-trained models to be compiled into static, high-performance C++ execution graphs within Simulink, achieving the sub-cycle inference speeds required for transmission-class protection.

5.2.1 Exporting PyTorch to Stateless ONNX Format

The trained .pth checkpoint is transitioned to the Simulink environment using Opset 14, a version optimized for Recurrent Neural Networks (RNNs). To maximize real-time determinism, the model is exported for "Stateless" inference. By utilizing a 32-sample sliding window (one full 50Hz cycle) to provide temporal context, the model does not need to persist hidden states (h_t) between execution steps, eliminating "hidden state amnesia" and cumulative error risks.

The export is configured with the following parameters:

- **Dynamic Axes:** Enabled for 'sequence_length' to allow for variable window sizes during experimental stress testing.
- **Constant Folding:** Enabled to prune the computational graph and reduce inference latency.
- **Input/Output Naming:** Specific identifiers ('dwt_features', 'trip_confidence') are utilized to facilitate direct HIL wiring.

5.2.2 Deep Learning Predict Block and Output Confidence Score

The ONNX model is integrated via the Simulink "Predict" block. During each 1600Hz simulation step, the 1×6 feature vector (Approximation and Detail energies for Phases A, B, and C) is buffered and reshaped into a [1, 32, 6] tensor. This 20ms "memory" of the spectral energy allows the LSTM to distinguish between the decaying harmonic signature of a magnetizing inrush and the sustained broadband energy of an internal fault.

The block outputs a floating-point confidence score ($P_{Internal}$), representing the AI's mathematical certainty of a fault signature. A score of 0.95 indicates a 95% probability of an internal fault, providing a nuanced metric that transcends simple binary thresholds. This digital confidence is then synchronized with the 87T RMS-based calculations for the final hybrid handshake.

5.3 The Hybrid Decision Engine

The protection system operates on a "Two-Key" security protocol: the physical operate conditions of the 87T logic and the pattern recognition of the LSTM must align before a trip command is sanctioned. This architecture recognizes that while AI is adept at waveform classification, the classical laws of physics provide the necessary safety net for power system reliability.

5.3.1 Adaptive 87T Logic vs. LSTM Confidence Handshake

The LSTM acts as a supervisory "Veto" logic. In scenarios involving deep CT saturation (typically occurring at 0.8 p.u. flux) or magnetizing inrush, the classical 87T logic may be "fooled" into its operate region due to distorted current waveforms. The LSTM, however, evaluates the "spectral health" via Detail energy (E_d), which specifically flags the discontinuities unique to faults. If the LSTM identifies an inrush signature or saturation-induced distortion, it issues a veto to block the 87T relay, maintaining 100% precision and preventing unnecessary blackouts.

5.3.2 Trip Truth Table and Operating States

The following table defines the decision-making matrix for the Hybrid Decision Engine:

Operating Mode	87T Region (OP)	ML Confidence (P)	Final Action
Magnetizing Inrush	1	High (≥ 0.85)	**BLOCK**

CT Saturation (Ext)	1	Med/High	**RESTRAIN**
Internal Fault	1	High (≥ 0.85)	**TRIP**
Internal Fault	1	Med ($0.60 \le P < 0.85$)	**HOLD/WAIT**
Internal Fault	0	Low	**RESTRAIN**
Normal Load	0	Low	**RESTRAIN**

5.3.3 Supervisory Override and Sensitivity Limits

To ensure survivability during catastrophic events, an **Instantaneous High-Set** ($I_{HS} > 10.0$ p.u.) bypasses the AI and the adaptive logic entirely. Such faults require immediate clearing to prevent structural damage to the 300MVA transformer. While the system is exceptionally secure, stress testing reveals a physical sensitivity limit: for High Impedance Faults (HIF) exceeding 50 Ω, the Detail energy signature diminishes, leading to a drop in Recall to approximately 69%. In these ambiguous cases where $P_{Internal}$ falls below 0.85, the relay defaults to a "Safety Net" fallback, adopting the conservative settings of the adaptive Springer logic.

5.3.4 Breaker Actuation and Anti-Pumping Circuit

The final trip command is processed through an **S-R Latch (Seal-In)** and an **Anti-Pumping** circuit. The S-R Latch ensures the trip command remains asserted until the breaker fully clears the fault, even if the initiating signal fluctuates as the arc is extinguished. The Anti-Pumping logic is critical for 300MVA infrastructure, preventing the breaker from repeatedly re-closing on a permanent fault if the "Close" command is inadvertently held, thereby protecting the breaker's mechanical integrity and preventing SF6 contact degradation.

This hybrid architecture transforms transformer protection from a fixed-threshold problem into an adaptive response system. By leveraging the 2ms inference speed of the LSTM—negligible compared to the 40-60ms mechanical delay of SF6 breakers—the relay achieves a superior balance of sub-cycle clearing speed and absolute security against benign transients.

Performance Evaluation and System Validation of a Hybrid DWT-LSTM Transformer Protection Scheme

6. RESULTS AND DISCUSSION

6.1 Offline LSTM Classification Performance

The strategic importance of baseline performance metrics is foundational in validating any novel protection architecture. Offline validation provides a strictly controlled environment to verify the core capability of the Long Short-Term Memory (LSTM) network to distinguish between classes based on features generated by the Discrete Wavelet Transform (DWT). In this stage, the objective is to confirm that the DWT (specifically the db4 mother wavelet) has successfully decoupled raw transients into a "linearly separable feature space." This pre-processing essentially simplifies the LSTM's task from solving a raw physics problem to a high-confidence pattern recognition success, establishing clear decision boundaries before the system faces the non-ideal complexities of a physical substation.

6.1.1 Accuracy, Precision, Recall, and F1-Score Evaluation

The model's performance was rigorously evaluated using an 80/20 train-test split on a dataset of 1,000 samples. The baseline results demonstrated near-perfect metrics, serving as a "proof of feature robustness" due to the DWT energy decoupling. To move beyond the risk of a "lucky" split and verify academic rigor, 5-fold cross-validation was employed.

Metric	Baseline (Test Set)	K-Fold Cross-Validation (Fold 1)
Accuracy	100.00%	94.37%
Precision	100.00%	96.20% (avg)
Recall	100.00%	88.89%

F1-Score	1.0000	0.9240 (avg)

The contrast between the 100% baseline and the 94.37% result in Fold 1 is technically instructive. While the DWT provides highly discriminative features, certain "hard cases"—specifically those involving marginal inception angles or deep saturation—introduce variance that requires the LSTM to generalize beyond simple memorization.

6.1.2 Confusion Matrix Analysis

The distillation of the baseline confusion matrix reveals the following distribution:

- **True Negatives (TN):** 113
- **True Positives (TP):** 87
- **False Positives (FP):** 0
- **False Negatives (FN):** 0

In power systems engineering, the significance of zero False Negatives (FN) is a metric of "Dependability." For a 300 MVA transformer, a failure to trip during a genuine internal fault does not merely represent an asset risk; it is a risk of catastrophic equipment failure and grid-wide instability. While these baseline results are ideal, the true utility of a relay-grade AI is defined by its performance under "Out-of-Distribution" stress conditions.

6.2 Stress Testing and Out-of-Distribution Robustness

To overcome the "Synthetic Data Paradox"—where models excel on clean, simulated data but fail in the field—the system was exposed to "Stress Test" datasets. These include High Impedance Faults (HIF), extreme noise, and deep CT saturation scenarios that were intentionally excluded from the primary training distribution. This identifies the "operating limits" of the system when faced with the "rude" noise of a physical substation environment.

6.2.1 Sensitivity Limitations during High Impedance Faults (HIF)

During stress testing, the model's Recall declined to 69.78%. A critical correlation exists between Fault Resistance (R_f) and detection capability:

1. **Standard Range (0–20 \Omega):** The model maintained near-perfect detection as these faults produce distinct energy spikes in the DWT detail coefficients.
2. **The Breaking Point (45–60 \Omega):** This represents the physical limit of the current architecture. Beyond this range, the wavelet energy signature "fades" into the load profile.

3. **HIF Range (50–100 \Omega):** Because the fault current is so minute, the AI classifies these internal events as "Normal Load," identifying a clear sensitivity ceiling that requires further data augmentation.

6.2.2 Security Validations during Extreme Noise and Deep CT Saturation

Despite the sensitivity trade-offs in HIF scenarios, the model demonstrated remarkable "Security." Even when subjected to 15 dB noise levels and 0.8 p.u. Remanent Flux (causing deep CT saturation), the Precision remained at 1.0000.

While training involved "polite" Gaussian noise, the stress test utilized "rude" noise profiles—mimicking lightning transients and DC battery ripples. The 1.0000 Precision confirms that the relay is "Safe and Secure"; it will effectively distinguish "Saturation Distortion" from "Fault Distortion," ensuring that external transients never cause a false blackout. For a protection engineer, this prioritizing of grid stability is a "Relay-Grade" achievement.

6.3 Real-Time Hybrid Relay Validation

The "Hybrid" architecture is implemented as a "Two-Key" launch protocol. The AI acts as a "Supervisory Veto" over the classical 87T Adaptive logic. In this configuration, the 87T provides the physical basis of Kirchhoff’s laws, while the LSTM provides the morphology-based intelligence to interpret the waveform's spectral health.

6.3.1 Prevention of False Trips (Sympathetic Inrush and External Faults)

The primary solution to the "False Trip" problem in large 300 MVA class transformers is this Supervisory Veto logic. During an external fault with deep CT saturation, a standard 87T relay may observe a "missing" current and flag an "Operate" condition. In the Hybrid Relay, the LSTM analyzes the jagged DWT coefficients and identifies them as saturation rather than an internal fault. The resulting low confidence score triggers a 'BLOCK' command, preventing a costly and unnecessary disconnection of a critical asset.

6.3.2 System Clearing Time and Computational Latency Benchmarks

For viability in a 50 Hz grid, the system must operate within strict timing constraints. The latency budget is analyzed as follows:

- **Data Buffering:** 5 ms (Quarter-cycle window for DWT extraction).
- **ONNX Inference:** ~1–2 ms (The "thought time" of the AI).
- **Breaker Mechanical Delay:** 40–60 ms (Physical contact separation).

With a total computational overhead of ~7 ms, the AI is effectively "thinking faster than the steel can move." This statistically insignificant latency confirms the hybrid system can clear internal faults well within sub-cycle requirements.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This research proves that the DWT-LSTM hybrid scheme provides a superior analysis of "waveform morphology" compared to traditional second-harmonic restraint. By moving beyond fixed harmonic ratios, the system adapts more effectively to the nonlinearities of modern smart grids.

7.1.1 Feature Robustness and AI Efficacy

The Daubechies-4 (db4) wavelet proved essential in simplifying the classification task. By extracting "Approx" and "Detail" energy from a 16-sample sliding window, raw transients were converted into linearly separable data points. This extraction allowed the LSTM to achieve high confidence scores with minimal latency, successfully converting a complex physics problem into a pattern recognition victory.

7.1.2 Enhanced Security of Hybrid Architecture

The "Security-First" design philosophy resulted in 100% Precision during extreme stress testing. This makes the model "Relay-Grade" and defendable to protection engineers, as it guarantees the AI will not introduce new risks of maloperation. Furthermore, the inclusion of an instantaneous overcurrent override ensures that even if the AI is "hesitant" due to low confidence, catastrophic bolted faults are still cleared by the physical high-set element.

7.2 Future Scope

While the current model excels at differential protection, comprehensive transformer health monitoring requires further specialization.

7.2.1 Exploration of Inter-Turn Winding Faults

A significant challenge remains in detecting minor inter-turn faults (e.g., 1–5% of turns), which often produce current shifts too small for standard 87T logic. Future work will utilize higher-resolution wavelet decomposition to identify the microscopic energy shifts associated with these insulation failures, meeting Smart Grid requirements to detect faults before they evolve into major phase-to-ground events.

7.2.2 Data Augmentation for High Impedance Fault Sensitivity

To address the 69.8% Recall limitation, "Data Augmentation" is proposed. By retraining the LSTM on a dedicated HIF dataset (20–100 \Omega), the model’s "vocabulary" can be expanded to include low-energy fault signatures. This will refine the decision boundary, maintaining security while significantly extending the relay's sensitivity.

The integration of Hybrid AI into the relay chassis marks a definitive step toward the evolution of a more resilient, self-aware, and autonomous modern Smart Grid.